Exercise Sheet 5: Inner Product Spaces

November 2024

In Exercises 1-2, find (a) $||u||$, (b) $||v||$, (c) $\langle u, v\rangle$, (d) $d(u, v)$ in the given inner product defined in $\mathbb{R}^n$.

1. $u = (1, -1, 2)$, $v = (2, 3, 1)$, $\langle u, v\rangle = u \cdot v$.
2. $u = (1, 1, 1)$, $v = (2, 5, 2)$, $\langle u, v\rangle = u_1v_1 + 2u_2v_2 + u_3v_3$.
3. Let $f(x) = x^2$ and $g(x) = x^2 + 1$ be functions in $C[-1, 1]$. Find (a) $\langle f, g\rangle$, (b) $||f||$, (c) $||g||$ for the inner product $\langle f, g\rangle = \int_{-1}^{1} f(x)g(x)dx$.
4. Determine whether $\langle u, v\rangle$ is an inner product for $u = (u_1, u_2)$, $v = (v_1, v_2)$, where:
   (a) $\langle u, v\rangle = u_1v_1 + 2u_2v_2$;
   (b) $\langle u, v\rangle = u_1v_1 - 2u_2v_2$;
   (c) $\langle u, v\rangle = u_1u_2 + v_1v_2$;
   (d) $\langle u, v\rangle = u_1^2v_1^2 + u_2^2v_2^2$.
5. Find the angle between $u = (1, 0, -3, 0)$ and $v = (2, -2, 1, 1)$ in the Euclidean 4-space $\mathbb{R}^4$.
6. Let $u = (1, 2)$, $v = (2, 5)$. Use the Euclidean inner product to find $\text{proj}_v u$.
7. Let $u = (1, -2, 2, 1)$. Find all vectors orthogonal to u in the Euclidean 4-space $\mathbb{R}^4$.

In Exercises 8-10, use the Gram–Schmidt orthonormalization process to transform the basis into an orthonormal basis. (Use the euclidean inner product and the vectors in the order in which they are shown.)

8. $B = \{(1, 1), (0, 1)\}$.
9. $B = \{(1, -2, 2), (2, 2, 1), (2, -1, -2)\}$.
10. $B = \{(3, 4, 0, 0), (-1, 1, 0, 0), (2, 1, 0, -1), (0, 1, 1, 0)\}$.
11. Find an orthonormal basis for the following subspace of Euclidean 3-space:
$$W = \{(x, y, z) \colon x + y + z = 0\}.$$
12. Find an orthonormal basis for the solution space of the homogeneous system of linear equations
$$x + y - z + w = 0$$
$$2x - y + z + 2w = 0.$$

13. Find the orthogonal complement $W^{\perp}$ of the subspace W of $\mathbb{R}^3$ spanned by the two column vectors of the matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 0 & -1 \end{bmatrix}$.

14. Find the projection of the vector onto the $v = (1, 0, -2)$ onto the subspace

$$W = \text{span}\{(0, -1, 1), (0, 1, 1)\}.$$

15. Find the least squares regression line for the set of data points $\{(-2, 2), (-1, 1), (0, 1), (1, 3)\}$. Graph the points and the line on the same set of axes.

16. Let $B = \{(0, 2, -2), (1, 0, -2)\}$ be a basis for a subspace of $\mathbb{R}^3$, and let $x = (-1, 4, 2)$ be a vector in the subspace.

    (a) Write x as a linear combination of the vectors in B. That is, find the coordinates of x relative to B.

    (b) Use the Gram–Schmidt orthonormalization process to transform B into an orthonormal basis B'.

    (c) Write x as a linear combination of the vectors in B'. That is, find the coordinates of x relative to B'.